

QE-CONVERSE user's manual

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1 Introduction

The QE-CONVERSE implement a non-perturbative approach (converse) to compute the orbital magnetization in isolated and periodic systems. The calculation of orbital magnetization allows ab-initio computation of macroscopic properties like the Nuclear Magnetic Resonance (NMR) chemical shifts and the Electronic Paramagnetic Resonance (EPR) g tensor.

2 Features

- Periodic and isolated systems
- Norm-conserving pseudopotentials
- Parallelization over bands and g-vectors
- NMR shielding tensors
- EPR g-tensor
- Electric field gradient (EFG) tensors and NMR/NQR quadrupolar parameters
- LDA and GGA functionals

3 Author contributions

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4 Build instructions

This current version of the code is compatible with the version 7.5 of Quantum-Espresso package.

1. A Quantum-Espresso package version 7.5 must be previously installed (<https://gitlab.com/QEF/q-e/-/releases/qe-7.5>). To take advantage of the enhancements in linear algebra operations, the configuration with scaLAPACK package or ELPA library is suggested. QE user's guide at http://www.quantum-espresso.org/user_guide/user_guide.html
2. Type:

```
git clone https://github.com/mammasmias/QE-CONVERSE
```

This will download from github the latest stable version of QE-CONVERSE.

3. Type:

```
cd QE-CONVERSE
```

4. Copy the source files and the Makefile from `/src/` directory into `/PP/` directory of Quantum-Espresso.
5. In the main directory of QE-7.5 type `make pp`. You should find the binary files `qe-converse.x` (NMR/EPR) and `qe-efg.x` (EFG) in the `/bin/` directory of Quantum-Espresso.

5 Quick start

To calculate NMR/EPR parameters you need:

1. pseudopotentials containing the GIPAW reconstruction (<https://sites.google.com/site/dceresoli/pseudopotentials>)
2. run `pw.x` to perform the SCF calculation
3. run `qe-converse.x` to calculate parameters (look into folder `examples` for NMR shielding, EPR g-tensor.)

6 Input file description

The input file consists on only one namelist `&input_qeconverse` with the following keywords:

prefix (type: character, default: `'prefix'`)

Description: prefix of files saved by program `pw.x`. The value of this keyword must be the same used in the SCF calculation.

outdir (type: character, default: `'./'`)

Description: temporary directory for `pw.x` restart files. The value of this keyword must be the same used in the SCF calculation.

diagonalization (type: string, default: `'david'`)

Description: diagonalization method (only allowed values: `'david'`)

verbosity (type: string default: `'high'`)

Description: verbosity level (allowed values: `'low'`, `'medium'`, `'high'`)

q_gipaw (type: real, default: 0.01, units: bohrradius⁻¹)

Description: the small wave-vector for the covariant finite difference formula.

dudk_method (type: string, default: `'covariant'`)

Description: k-point derivative method (only allowed values: `'covariant'`)

diag_thr_init (type: real, default: 10⁻⁷, units: Ry²)

Description: Convergence threshold (ethr) for iterative diagonalization.

conv_threshold (type: real, default: 10⁻⁸, units: Ry²)

Description: convergence threshold for the diagonalization in the SCF step.

mixing_beta (type: real, default: 0.5)

Description: mixing factor for self-consistency.

lambda_so(1,...,3) (type: real, default: 0.0, units: Bohr magneton)

Description: Cartesian components of electron spin. The value (1,...,3) denotes the spin-orbit coupling direction.

m_0(1,...,3) (type: real, default: 0.0, units: nuclear magneton)

Description: Cartesian components of nuclear dipole. The value (1,...,3) denotes the nuclear dipole moment direction.

m_0_atom (type: integer, default: 0)

Description: Atom index carrying the nuclear magnetic dipole.

delete_dudk_files (type: logical, default: `.false.`)

Description: If `.true.`, temporary du/dk files (wavefunction k-derivatives) are deleted after the orbital magnetization calculation.

dudk_in_memory (type: logical, default: `.false.`)

Description: If `.true.`, du/dk gradients (wavefunction k-derivatives) are stored in memory rather than written to disk. This is useful when there is enough memory to store these gradients.

7 Electric field gradient (EFG)

The electric field gradient at each nucleus is a ground-state property and is computed by a separate binary, `qe-efg.x`, directly from the converged SCF charge density and wavefunctions (no converse SCF is needed). A single ordinary `pw.x` run is sufficient; symmetry may be left on. The total EFG is the sum of the bare valence, ionic (Ewald), and GIPAW reconstruction contributions, which are printed separately together with the principal values and the quadrupolar parameters C_q , η , and ν_Q .

The input is the namelist `&input_qeefg`:

prefix (character) prefix used in the SCF `pw.x` calculation.

outdir (character) scratch directory used in the SCF `pw.x` calculation.

q_efg(n) (real, default 0) nuclear quadrupole moment of atom type n , in units of 10^{-30} m^2 (= 10 mbarn); C_q is printed only for types with a non-zero value.

i_efg(n) (real, default 0) nuclear spin I of atom type n ; the quadrupolar frequency ν_Q is printed only for types with $I \geq 1$.

Atom types are indexed as in the `ATOMIC_SPECIES` block of the `pw.x` input. Run `pw.x` followed by `qe-efg.x` to obtain the EFG at every site.

8 Limitations

K.POINTS GAMMA are not supported. For Gamma-point calculations use: K.POINTS automatic and ‘1 1 1 0 0 0’

9 Resources

- NMR periodic table: <http://www.pascal-man.com/periodic-table/periodictable.html>

References

[1]